When Computers Were Human

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FOR JEAN

Who took my people to be her people and my stories to be her own without realizing that she would have to accept a comet, the WPA, and the oft-told tale of a forgotten grandmother

CHAPTER TWO

The Children of Adam Smith

> Even in the quieter professions, there is a toil and a labour of the mind, if not of the body. . . .
>
> Jane Austen, *Persuasion* (1818)

"THE SUPERIOR GENIUS and sagacity of Sir Isaac Newton," wrote the philosopher Adam Smith, "made the most happy, and, we may now say, the greatest and most admirable improvement that was ever made in philosophy." Smith turned his generous praise on Newton's calculus and stated that new discoveries would come from "more laborious and accurate calculations from these principles."[1] At least one scholar has found Smith's praise insincere and has suggested that the philosopher distrusted any science that rested "primarily upon mathematics, rather than easily visualized phenomena, common to the mind of all men."[2] Smith was more interested in things of earth than in things of heaven, the movement of goods and services rather than the cycles of planets and comets. At the time of the 1758 return, he was collecting the material that formed the basis for his book *An Inquiry into the Nature and Causes of the Wealth of Nations*. In this work, he sought a fixed principle that explained economic behavior as well as Newton's gravity explained celestial motion. He found this principle in the marketplace, "the propensity to truck, barter, and exchange one thing for another," though he was not certain that the rules of the market equaled those of celestial motion. "Whether this propensity be one of those original principles in human nature," he speculated, "or whether, as seems more probable, it be the necessary consequence of the faculties of reason and speech, it belongs not to our present subject to inquire. It is common to all men, and is to be found in no other race of animals."[3]

Smith claimed that the market encouraged people to specialize, to produce those goods that gained them the most profit. Butchers did not make shoes, nor did cobblers slaughter their own animals. This specialization was one part of a more general idea that Smith identified as the division of labor. "The greatest improvements in the productive powers of labour," he wrote, "seem to have been the effects of the division of labour." He claimed that there were three benefits to be gained from such division. First, it led to the "increase of dexterity in every particular

workman." Laborers could focus on a small number of tasks and thus gain skill and efficiency. Second, divided labor made workers more productive by reducing "the time which is commonly lost in passing from one species of work to another." Finally, the division of labor encouraged workers to improve their tools, to invent "a great number of machines which facilitate and abridge labour, and enable one man to do the work of many."[4]

Later generations would explore Smith's market principles and divided labor with the calculus of Isaac Newton. Smith was content to simply describe how his laws touched different aspects of economic behavior. He claimed that his ideas applied equally to manufacture and to "natural philosophy," the term he used to describe scientific research. The "subdivision of employment in philosophy," he wrote, "improves dexterity, and saves time." Smith conceded that philosophers might not be motivated by the traditional economic forces of profit and loss, yet he argued that they desired to extract the greatest results from their limited resources, so that "more work is done upon the whole, and the quantity of science is considerably increased by it."[5] The work of Alexis Clairaut, Nicole-Reine Lepaute, and Joseph Lalande was an early example of this observation. Without the division of labor, Clairaut could not have completed the calculations before the comet's reappearance and could not have devoted so much effort to checking the results.

At the time that Adam Smith was writing *The Wealth of Nations*, the British Admiralty, the executive office of the English navy, was organizing a new computing office and taking a further step in the division of labor. The Admiralty created this office in order to produce a nautical almanac, a volume of tables that gave the position of the sun and the moon, the planets and the stars. The founder of this office was the new Astronomer Royal, Nevil Maskelyne (1732–1811). Maskelyne was the successor, twice removed, of Edmund Halley, the fifth scientist to oversee the Royal Observatory at Greenwich. This appointment was not purely a scientific honor, as it carried a practical responsibility for the country's fleet of naval and merchant ships. Anyone who accepted the king's warrant for astronomy was required to develop methods of celestial navigation, particularly techniques for the "finding out of the longitude of places."[6] As if to emphasize this charge, the Greenwich Observatory sat on the high bank of the River Thames in the midst of a royal estate. From his desk, Maskelyne could view the ocean traffic as it moved between the London docks and the open waters of the North Sea.

The *Nautical Almanac* was the outgrowth of a competition between two methods for finding longitude, one computational and the other mechanical. The two methods were nearly identical and differed only on a

single point: the means of determining the time at Greenwich. The time at Greenwich was important because it allowed a navigator to compare two observations of a single star. The first measurement would be taken by the navigator in the dim moments before dawn or in the dusky hour of twilight, when the thin line of the horizon was visible from the ship and at least a few bright stars could be seen in the violet sky. After determining the position of a star, the navigator would turn to a nautical almanac and find the position of the same star as it would be viewed at the identical moment from the observatory at Greenwich. The difference between these two positions, properly adjusted with a dozen steps of calculation, was the longitude of the ship.

In the 1760s, there were two possible ways of determining the time at Greenwich, both with advantages and drawbacks. The simpler way used a mechanical clock set to the time at Greenwich. This solution was problematic, as no common clock could guarantee sufficient precision under shipboard conditions. The roll of the waves disrupted pendulums. Variations in heat and humidity caused springs to expand and contract. A good clock might lose or gain four minutes a day, enough time to allow the earth to spin a full degree in its rotation. In the middle latitudes, a four-minute error could translate into a deviation of fifty miles. A navigator relying on such a clock could easily calculate a longitude that placed his ship at a safe distance from the shore when, in fact, the vessel was about to strike coastal rocks. In the early 1760s, English inventors strove to develop a precision clock that could record the time accurately under shipboard conditions. Of the timekeeping devices presented to the British Admiralty, one created by John Harrison (1693–1776) was the most promising.[7]

The second approach to determining the time at Greenwich used the moon as a timekeeper. This technique was known as the lunar distance method. The moon moves twelve degrees across the sky each night, passing neighboring stars as if they were marks on a watch dial. That motion is enough to allow a skilled navigator to compute the time at Greenwich with sufficient accuracy, though the calculations are admittedly lengthy and require a special table that predicts the moon's position. The lunar distance method had been developed in the early eighteenth century but had been dismissed by most navigators because of the difficulties in predicting the position of the moon. Like the calculation of the perihelion for Halley's comet, the prediction of lunar position required the solution of a three-body problem. In this case, the three-body system involved the moon, the earth, and the sun. An acceptable solution to this particular system appeared only in the late 1750s, when the German astronomer Tobias Mayer (1723–1762) published a detailed table of lunar positions.[8] Astronomers praised Mayer's work as "the most admirable masterpiece in theoretical astronomy," and in 1761, the *Connaissance des Temps* published an article that showed how Mayer's tables could be used in navigation.[9]

In popular accounts of the competition between Harrison's clock and the lunar distance method, Nevil Maskelyne has been portrayed as a villain, a powerful scientist who undercut a valid technology for personal reasons. His alleged villainy came when he was asked by the British Admiralty to compare Harrison's clock with the lunar distance method. Some writers have charged that Maskelyne was a prejudiced evaluator of the two techniques because he had publicly stated his admiration of Mayer's lunar tables before the trial began and was known to favor the techniques of astronomical calculation.[10] His conclusions from a test voyage certainly confirmed his opinion that the lunar distance method was a practicable means of determining longitude, and he dismissed Harrison's clock.[11] From a modern perspective, precision clocks, now called chronometers, clearly provide the easiest way of determining the time at Greenwich, but such a conclusion may not have been so clear in the 1760s. The historian Mary Croarken has noted that Harrison's clock was an immature technology and was "much too expensive to be taken to sea by the majority of [English] navigators."[12]

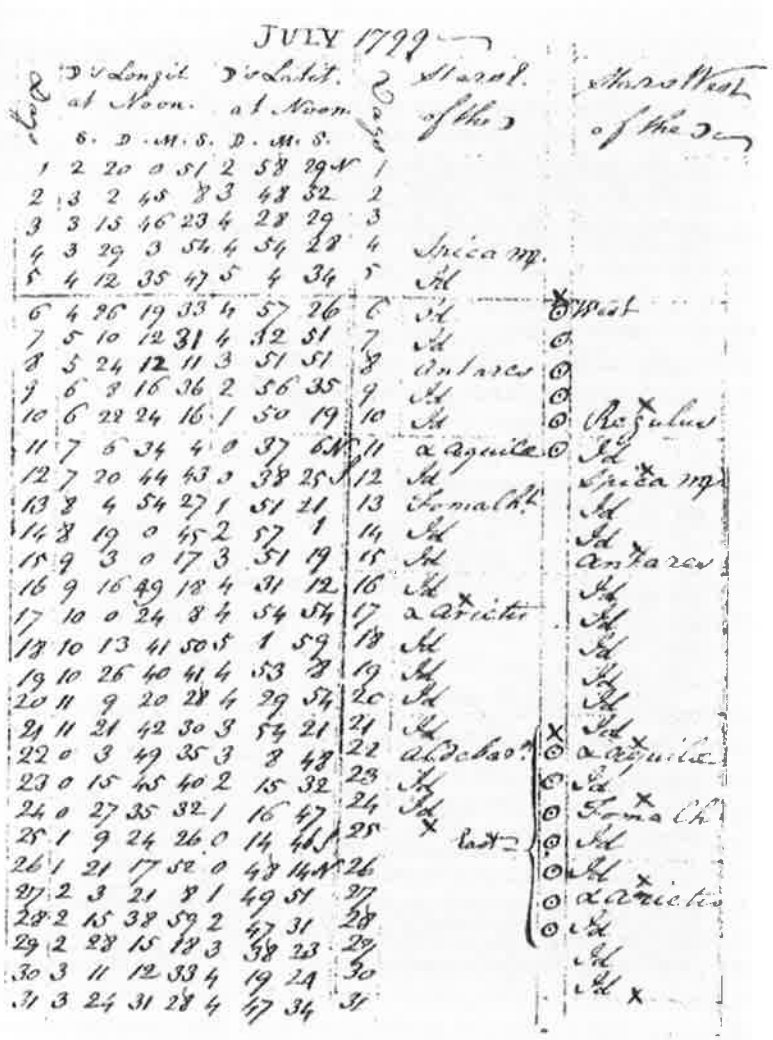
On the trial voyage from England to Barbados and back, Maskelyne had required four hours to make a single computation of longitude with the lunar distance method.[13] "It is rather to be wished," he wrote, "that such parts of the computations as conveniently can, were made previously at land by capable persons."[14] Those parts of the computations that could be done in advance took the form of a set of tables that gave the distance from the moon to easily recognizable stars in a simplified form. These tables needed to be prepared and published annually, as the position of the moon varied from year to year. With such tables, a navigator could compute the time at Greenwich with a handful of operations and determine a ship's longitude with only thirty minutes of work.[15] Maskelyne wanted to include these tables as part of a general nautical almanac, as such values could be used for purposes beyond the problem of finding the time at Greenwich. They could even be used to check the settings of a chronometer in the middle of the ocean or guide a ship back to land should the chronometer fail.

In February 1765, the British Admiralty approved Maskelyne's plan for an almanac, gave him a staff of five computers, and told him to begin work on celestial tables for 1767. To all involved with the project, including Maskelyne and the members of the Admiralty, this assignment must have seemed quite reasonable. Under normal circumstances, the almanac staff would have to produce a new publication every year. Maskelyne had fully twenty-two months before the start of 1767, almost twice

the time he should have needed, but the process of recruiting and training his computing staff proved to be more challenging than he had anticipated. He organized the computing staff as a cottage industry, a form of production that was still common in England, even though it was starting to be eclipsed by the factory. In cottage production, the workers labored in their own homes. Their materials, instructions, and often their tools were provided by the company or individual for whom they worked. In the clothing industry, cottage workers might receive carded wool and spin it into yarn. For the *Nautical Almanac* computers, Maskelyne provided paper, ink, and instructions that were called "computing plans." Maskelyne wrote these plans on one side of a heavy sheet of folded stationery. The instructions, scrawled in a slightly disheveled hand, summarized each step of the calculation. Occasionally, he would illustrate the computations with a hasty sketch of an astronomical triangle. On the other side of the paper he drew a blank table, ready for the computer to complete.[16]

The computers produced tables that tracked the motion of a planet or the sun, tables that were called ephemerides in the plural (or an ephemeris in the singular). Most of these ephemerides were double-computed, prepared by two independent computers working from the same plan. Each computer would send Maskelyne a version of the ephemeris. Maskelyne would forward the two ephemerides to a third computer, who had the title of comparator. The comparator would search the two ephemerides for discrepancies and correct the mistakes. The only tables that were not double-computed were those of lunar motion. These tables were divided in half. One computer would calculate the moon's position at noon. The other would compute the position at midnight. The comparator would merge the two tables and make sure that the two sets of calculations were consistent.[17]

Initially, Maskelyne assigned two computers to prepare the 1767 volume of the almanac. A third acted as the comparator, and the remaining two were put to work on the 1768 volume. From what we know of his staff, all of them came from the second tier of astronomical talent. Most commonly, they had demonstrated some skill at astronomy but lacked the resources or the connections to acquire one of the prestigious scientific appointments at Cambridge or Oxford. The first computer of the 1767 volume, William Wales (1734–1798), came from a poor family in the north of England. The second computer, Israel Lyons (1739–1775), was a Jew and was unwilling to make the profession of belief that might gain him a place at the church-centered universities. The comparator, Richard Dunthorne (1711–1775), had shown the greatest ability to advance himself as a scientist. He, too, was born to a lower-class family but had demonstrated his mathematical prowess by analyzing the motion of



4a. Computing sheet of Nevil Maskelyne

the moon. This work had given him a minor reputation as an astronomer and had connected him to a wealthy patron who provided Dunthorne with a regular income.[18]

Whenever possible, Maskelyne attempted to reduce the amount of calculation by borrowing tables from other sources, such as the *Connaissance des Temps*. He also simplified some of the calculations by employing the method of interpolation. Interpolation expands a table by esti-

4b. Computing instructions prepared by Nevil Maskelyne

mating intermediate values rather than by calculating these numbers from the original equations. It is a mathematical means of connecting the dots. The computers would link the moon's position at noon to its location at midnight using a polynomial, a mathematical expression that is the sum of terms such as x, x^2, and x^3. With this polynomial, the computers estimated the moon's location at three-hour intervals without having to calculate new values from Mayer's tables.[19]

Even with Maskelyne's attempts to minimize the workload, the 1767

almanac fell behind schedule. Wales and Lyons needed time to learn the new computing procedures and develop the skill that would get the work done most efficiently. By the spring of 1766, Maskelyne recognized that his two computers would not be able to finish their computations in time for publication the following fall. Both were engaged in at least one other job and could not devote extra time to completing the calculations. Lyons worked as a surveyor and did other work for the British Admiralty, while Wales was involved in a number of astronomical projects. Fortunately, Maskelyne had a reserve pool of labor, the two computers who were preparing the second issue of the almanac. He told this pair to put aside their calculations and assist Wales, Lyons, and Dunthorne with the first issue. Together, the five computers finished the tables in late fall. The 1767 issue of the almanac appeared only six days after the start of the year.[20]

As the computers moved to the second and third almanacs, they were able to claim the first benefit that Smith had ascribed to divided labor, the increase in dexterity and speed. After three years of calculations, the almanac staff had completed all of the almanacs through the 1773 volume and were beginning the calculations for 1774. By 1780, they were creating tables six years in advance, and Maskelyne was able to reduce the number of computers from five to four.[21]

The division of labor for Maskelyne's first *Nautical Almanac* offered no innovation beyond the methods commonly applied in English commerce. The only difference between the computers and the carders and weavers of the cloth industry was the fact that the computers' product, the ephemerides, could be folded into a neat packet and sent through the mails. A more radical approach to the division of labor was found at the French Bureau du Cadastre. The bureau, a civil mapping agency, was a product of the French Revolution and hence embraced notions about labor and organization that were far more radical than those employed by Maskelyne at the British *Nautical Almanac*. The bureau prepared maps for governance, taxation, and land transactions. Initially, it had no computing division beyond a few assistant surveyors. It assembled a staff of almost one hundred computers when it became involved with the standardization of weights and measures that produced the metric system.

The metric system grew out of an attempt by the National Assembly to gain control of the French economy. In March of 1790, barely eight months after the storming of the Bastille prison, the National Assembly debated a proposal to discard "the incalculable variety in our weights and measures and their bizarre names" and adopt a unified measurement system based upon scientific principles.[22] At that time, each region was free to establish its own set of measures. Local officials easily manipu-

lated these measures to their own advantage in a number of ways. Commonly, they could keep a large measure to collect taxes of grain and produce but reserve smaller measures for the payment of their own debts.[23]

The Académie des Sciences agreed to create the new system of weights and measures. They quickly stipulated several basic principles for the new system. They agreed that the standards of weight and length should be beyond the control of any political organization and that the units for area, volume, and even weight should be related to the unit for length. In one of their final discussions, the Académie stated that the new measures should form a decimal system. All units should be related through multiples of ten. For example, the meter, the standard measure of length, would be divided by ten to produce the decimeter, which in turn could be divided to produce the centimeter, the millimeter, and the micrometer. The liter, the gram, and the dyne, the standard units of volume, mass, and force, could also be divided or expanded in decimal multiples. The members of the Académie argued that this same principle should govern all standard units, including those that measured angles. Under their proposal, a right angle would no longer have 90 degrees. Instead, it would be split into one hundred new units called grades.[24]

The proposal for the decimal measurement of angles produced a major computational problem that led to the creation of a computing office at the Bureau du Cadastre. The principal users of angle measure, navigators and surveyors, did their work with sines, cosines, and other trigonometric functions. Without trigonometric tables prepared for the decimal grades, the new standard for angle measure would be unused. No surveyor or navigator, even one ardently committed to the revolutionary cause, would measure angles in grades if he had to convert his numbers into degrees in order to use a sine table. Openly or surreptitiously, they would measure their angles in degrees and use the trigonometric tables of the ancien régime to calculate their position on the globe or the area of a piece of land.

The director of the Bureau du Cadastre was Gaspard Clair François Marie Riche de Prony (1755–1839), a civil engineer with the country's elite Corps des Ponts et Chaussées, the Corps of Bridges and Highways. De Prony came from a family of "modest but ancient title" in the province of Beaujolais. His mathematical skill had brought him to the attention of the corps and gained him entrance to the corps's preparatory school in Paris. He graduated at the top of his class from the school and proudly accepted the uniform of a corps officer, which came with royal fleur-de-lys buttons.[25]

Following his graduation, de Prony had hoped to live in the capital and pursue scientific research, but the conventional assignment for young engineers was a term of service in the field. He fulfilled his duty and found

a way back to Paris by taking an appointment as an inspector, an officer who critiqued the mathematical analyses of other engineers and verified calculations. Though the assignment involved no independent work, he found it a pleasing activity, reporting that "happy circumstances then put me in contact, with the most distinguished savants of the capital." It also brought him into contact with foreign savants, including Nevil Maskelyne. De Prony met Maskelyne when the corps decided that it needed to remeasure the difference in longitude between Paris and London. Beyond the intellectual challenge of this work, an accurate measurement of the difference would allow French surveyors to use the British *Nautical Almanac* in their work and enable the Académie des Sciences computers to compare their tables with the output of Maskelyne's computers. As part of the effort to measure the difference between the two capitals, de Prony traveled to Greenwich, met Maskelyne, and inspected the work of the almanac computers.[26]

De Prony had become leader of the Bureau du Cadastre because of a cautious and uneasy relationship between the corps engineers and revolutionaries. The revolutionaries were wary of the corps, their uniforms, and their fleur-de-lys buttons, yet they needed the services of corps officers and could not easily dismember the group. In an attempt to weaken the authority of the corps, the revolutionary government tried to disperse the officers, most of whom resided in Paris, and have them take positions in the countryside.[27] In early 1791, the government ordered de Prony to move to southwest France but found that he was unwilling to take the assignment. "I have received your letter, informing me that I am appointed engineer-in-chief for the departement of the Pyrenees," he wrote to his superintendent. "In these circumstances, I beg you to permit me to remain in Paris." He argued that he was working on two reference books that would be difficult to complete without the resources found in the capital. Claiming that the research had "already cost me many sleepless nights," he stated that he was willing to forgo a formal assignment so long as he could stay in Paris and continue with his research. After a brief confrontation, a senior officer intervened and gave de Prony the job at the Bureau du Cadastre.[28]

In ordinary times, de Prony would have been responsible for organizing surveying teams and dispersing them across the country, but 1791 was no ordinary time. The dangers of civil unrest forced de Prony to keep the survey teams in Paris and occupy them with "tasks of public usefulness."[29] Under these circumstances, he sought or accepted the task of preparing the trigonometric tables for the decimal grade system of angle measure. De Prony wrote that the assignment "over-burdened" him.[30] The Académie des Sciences wanted the work done quickly and required that the final tables leave "nothing to be desired in accuracy."[31] As he

contemplated the work before him, he devised a plan based upon the first chapter of *The Wealth of Nations*. A French commentator has suggested that de Prony approached the book haphazardly, the way one might flip open a Bible to search for an inspiring verse. "He opened it at random and his eye fell on the first chapter, which was called *A Treatise on the Division of Labor*, and which cited the example of making pins."[32] It is a dramatic rendition of the story, but it overlooks the long-standing influence of *The Wealth of Nations* upon the engineers in the Corps des Ponts et Chaussées. The book had circulated among the officers during the early 1780s and was probably an old and familiar friend to de Prony.[33]

De Prony recalled that the pin example provided the insight he needed. "Suddenly," he wrote, "I conceived how I might apply the same method to the work which had burdened me." Referring to one of the tables he needed to compute, he recorded that his bureau "could manufacture logarithms as easily as one manufactures pins."[34] His recollection of the event ignores a great deal of hard work and gives his accomplishment a patina of false confidence. The structure of de Prony's computing office cannot be easily seen in Smith's example. His computing staff had two distinct classes of workers. The larger of these was a staff of nearly ninety computers. These workers were quite different from Smith's pin makers or even from the computers at the British *Nautical Almanac* and the *Connaissance des Temps*. Many of de Prony's computers were former servants or wig dressers, who had lost their jobs when the Revolution rendered the elegant styles of Louis XVI unfashionable or even treasonous.[35] They were not trained in mathematics and held no special interest in science. De Prony reported that most of them "had no knowledge of arithmetic beyond the two first rules [of addition and subtraction]."[36] They were little different from manual workers and could not discern whether they were computing trigonometric functions, logarithms, or the orbit of Halley's comet. One labor historian has described them as intellectual machines, "grasping and releasing a single piece of 'data' over and over again."[37]

The second class of workers prepared instructions for the computation and oversaw the actual calculations. De Prony had no special title for this group of workers, but subsequent computing organizations came to use the term "planning committee" or merely "planners," as they were the ones who actually planned the calculations. There were eight planners in de Prony's organization. Most of them were experienced computers who had worked for either the Bureau du Cadastre or the Paris Observatory. A few had made interesting contributions to mathematical theory, but the majority had dealt only with the problems of practical mathematics.[38] They took the basic equations for the trigonometric functions and reduced them to the fundamental operations of addition and subtraction.

From this reduction, they prepared worksheets for the computers. Unlike Nevil Maskelyne's worksheets, which gave general equations to the computers, these sheets identified every operation of the calculation and left nothing for the workers to interpret. Each step of the calculation was followed by a blank space for the computers to fill with a number. Each table required hundreds of these sheets, all identical except for a single unique starting value at the top of the page.

Once the computers had completed their sheets, they returned their results to the planners. The planners assembled the tables and checked the final values. The task of checking the results was a substantial burden in itself. The group did not double-compute, as that would have obviously doubled the workload. Instead the planners checked the final values by taking differences between adjacent values in order to identify miscalculated numbers. This procedure, known as "differencing," was an important innovation for human computers. As one observer noted, differencing removed the "necessity of repeating, or even of examining, the whole of the work done by the [computing] section."[39]

The entire operation was overseen by a handful of accomplished scientists, who "had little or nothing to do with the actual numerical work." This group included some of France's most accomplished mathematicians, such as Adrien-Marie Legendre (1752–1833) and Lazare-Nicolas-Marguerite Carnot (1753–1823).[40] These scientists researched the appropriate formulas for the calculations and identified potential problems. Each formula was an approximation, as no trigonometric function can be written as an exact combination of additions and subtractions. The mathematicians analyzed the quality of the approximations and verified that all the formulas produced values adequately close to the true values of the trigonometric functions.

Joseph Lalande visited the Bureau du Cadastre in 1794, after the computers had been working for nearly two years. He probably saw only the office of de Prony and his planners. No record has been found of a centralized computing floor for the former hairdressers.[41] Given the size of the computing staff and the tradition of cottage work, it is likely that the computers did their calculations at home. Lalande, who had become the éminence grise of French astronomy, reported that de Prony's computers were "producing seven hundred results each day."[42] At that pace, they could have duplicated Clairaut's calculations for Halley's comet in about a week. Perhaps more to the point, if Lalande, Lepaute, and Clairaut had been asked to prepare de Prony's decimal trigonometry tables, they would have spent a century and a half sitting at their table in the Palais Luxembourg.

Lalande's visit seems to have marked a high point of the cadastral computers. In 1795, the revolutionary government instructed de Prony to

prepare the tables for publication "at the expense of the nation,"[43] but by the time de Prony completed the work, the nation was not all that interested in paying for the new trigonometry. Decimal angle measure was not included in the law implementing the new metric system, which was passed by the National Assembly on August 1 of that year.[44] "In the face of popular indifference and hostility," wrote historian Ken Alder, "the government began to lower its sights."[45] Officials had difficulty enforcing the use of metric measures in Paris, and there is little evidence that it penetrated into the countryside.

De Prony kept his project in operation through 1800 or 1801, even though it appears that most of the work had been completed by 1796. Even before the typesetters began work on the tables, de Prony's publisher began to promote the new trigonometric functions with the story of their creation. According to the advertisement, de Prony had created a new "process of manufacturing" that was "strange in the history of science, where there is no other example." The publisher argued that the tables would never have been created "if M. de Prony had not had the fortunate idea of applying the powerful method of division of labour," but such words could not bring the tables into print.[46] The publisher was bankrupted during a national fiscal crisis. The French government, then led by Napoleon, had no interest in completing the work. De Prony retained the nineteen-volume manuscript and made occasional, though fruitless, efforts to publish it.[47]

The story of Gaspard de Prony and his computers at the Bureau du Cadastre would have been little more than an odd footnote to the history of economics were it not for the attention of Charles Babbage. Babbage is generally remembered as a mathematician and as a designer of early computing machinery, but he was a broad and eclectic scholar whose interests ranged from mathematics and astronomy to economics and railroad construction. To a certain degree, he fit the stereotype of the Victorian gentleman scientist. He had been educated at Cambridge in the canonical works of Newton and Halley; he possessed a comfortable, though not extravagant, income that allowed him to pursue his own interests; he lived in London and mingled freely with the country's political and intellectual elite. In some ways, he was more interested in the organizations and institutions of science than he was in the science itself.[48]

Babbage arrived in London in 1814, finished with his Cambridge studies and newly married. He applied to be a computer at the Royal Greenwich Observatory, but friends encouraged him to direct his talents elsewhere.[49] Looking for ways to establish a reputation as a scientist, Babbage decided to give a series of public lectures on astronomy. His mathematical training was deeply connected to astronomical problems,

5. Charles Babbage

but that training did not make him an expert on stars and comets and planets. In preparing the lectures, he relied on the advice of more accomplished colleagues, notably a college friend, John Herschel (1792–1871), and Herschel's aunt, Caroline Herschel (1750–1848). The two were members of England's premier astronomical family. William Herschel (1738–1822), the father of John and the brother of Caroline, had discovered the planet Uranus in 1781. Caroline Herschel had served as her brother's assistant before she became recognized for discovering comets and cataloguing nebulae.[50]

Babbage's lectures brought him to the attention of a group of businessmen and amateur astronomers who were organizing a society for "the encouragement and promotion of Astronomy." This group, originally called the Astronomical Society of London, invited Babbage and John Herschel to their first meeting. This meeting was held in January 1820 at a tavern situated among the business houses of central London.

Most of the founders had some connection to ocean trade and the problems of celestial navigation. They were merchants, currency traders, stockbrokers, and business teachers.[51] Though many of them were comfortable behind the lens of a telescope or computing the orbit of a planet, they described astronomy as if it were another commercial endeavor. The language of Adam Smith and divided labor permeated their words and sentences. They stated that the society would coordinate the "labours of insulated and independent individuals" and that they were "ready and desirous to divide at once the labour and the glory" of celestial observation. Their descriptions of the society suggest that they had some first-hand experience with the problems of management, for they wrote of the need "to preserve a perfect unity of design" while simultaneously preventing the loss of effort.[52]

In 1821, Babbage and Herschel agreed to undertake one of the first projects sponsored by the society, a set of mathematical tables that would augment the material in the British *Nautical Almanac*. The two college friends prepared a computing plan and hired a pair of computers to produce two independent versions of the table. Once the computers had finished their work, Babbage and Herschel compared the results. Sitting together, one of them read aloud the values from his version of the table. The other held the second table and confirmed each number. "Finding many discordancies," Babbage later wrote, "I expressed to my friend the wish that we could calculate by steam."[53] He would point to this work as the start of his study of computing machines, but he would repeat at least two inconsistent versions of this story. A second narrative placed his moment of insight during his student days in Cambridge. Both versions suggested that the idea to design a computing machine grew out of a desire to improve the accuracy of calculation. As Babbage would acknowledge, he was neither a trained engineer nor a skilled machinist.[54] His design for a computing engine would be based upon the division of labor, though not upon Maskelyne's ideas but upon those of de Prony.[55]

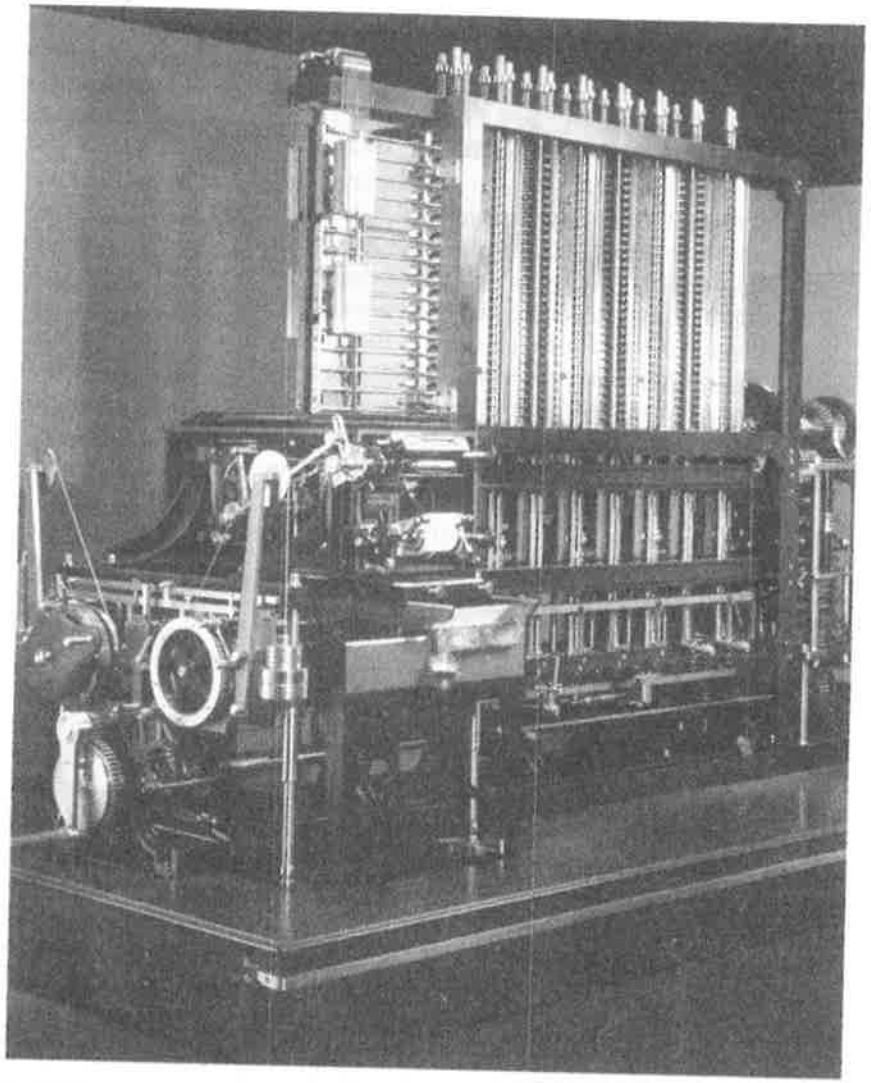
Babbage knew of de Prony and the cadastral computers when he and Herschel prepared their tables in 1821. He may have seen the manuscript tables when he visited Paris with his wife in 1819, or he may have learned of the computing effort when de Prony asked a wealthy English physician to help publish the tables.[56] Babbage was little interested in the decimal trigonometry tables but clearly understood the benefits of de Prony's organizational plan. He wrote that de Prony's experience showed that the division of labor was not restricted to physical work but could be applied to "some of the sublimest investigations of the human mind," including the work of calculation. After his own attempts at calculation, Babbage turned to de Prony's analysis and took the division of labor to its next logical step, the invention of a machine to "facilitate and abridge" the work.[57]

When he began work on his calculating machine, Babbage was following a path that was already well marked. Inventors in both England and the United States had built machines based upon Adam Smith's example of the divided labor in pin manufacture. Their machines followed each step that Smith had identified in *The Wealth of Nations*. They cut a roll of wire into fixed lengths, sharpened one end of each wire segment, affixed a head to the other, and placed the finished pin in a paper holder. Babbage took the opportunity to study one of these pin-making machines and reported that "it is highly ingenious in point of contrivance," especially interesting "in respect to its economical principles."[58]

Babbage designed a machine that might be considered more flexible than the pin-making machines. Rather than analyze the equations that had been used to create a specific table, such as the decimal trigonometry tables, he considered a single computational technique that could be applied to many kinds of calculation. The technique that he chose was a process of mathematical interpolation known as the finite difference method. The finite difference method is one way of computing intermediate values of a table, such as the intermediate positions of the moon that Nevil Maskelyne's computers prepared for the British *Nautical Almanac*. It is especially amenable to the division of labor because it reduces the entire process into the simple operation of addition. A simple application of this method can compute a list of the squared integers (4, 9, 16, etc.) without performing a single multiplication. First, one computes a list of the odd integers: 1, 3, 5, 7, 9, etc. This can be done by starting with the number 1 and successively adding 2: $1 + 2 = 3$, $3 + 2 = 5$, $5 + 2 = 7$. Once this has been completed, one can sum the list of odd integers to get the list of squares: $1 + 3 = 4$, $4 + 5 = 9$, $9 + 7 = 16$, and so on.

Though Babbage's machine would be far more complicated than the pin-making device, it was simpler in one respect. The pin machine needed to perform four different fundamental operations. Babbage's machine would only need to do one: addition. Babbage started with a geared adding mechanism originally developed by Blaise Pascal (1623–1662) in 1642,[59] improved the design, and cascaded the devices so that the results of one addition would be fed to the next. To create a list of squared integers, one mechanism would repeatedly sum the number 2 to create new odd numbers. The next mechanism would sum the odd numbers to create the squares. By the spring of 1822, Babbage had completed a demonstration model of his machine, which he named the "Difference Engine."

The London of 1822 was wholly unprepared for Babbage's machine. It was a world of gaslights and horse-drawn carriages, of servants and walking sticks. Most residents had not yet seen a steam locomotive, as the city's first railroads were still under construction.[60] Though the idea of the adding machine was one hundred and eighty years old, there was



6. Difference engine constructed from the original plans of Charles Babbage

none to be bought or sold. The first commercial machine, which would be produced in France, existed only as a crude prototype.[61] Babbage anticipated that his machine might be met with disbelief or even opposition. He cautiously approached the Royal Society, recognizing that the organization might be able to help him promote his machine. His letter to the society president made conservative claims and acknowledged that Royal Society members might not believe it possible to create a machine that

could handle such complex calculations without supervision. He tried to disarm potential criticism by invoking Jonathan Swift: "I am aware that the statements contained in this Letter may perhaps be viewed as something more than utopian, and that the philosophers of Laputa may be called up to dispute my claim to originality."[62]

The Laputian philosopher that Babbage wished to avoid was not the computing tailor but an inventor who lived on the flying island. This inventor claimed to have a computing machine whereby "the most ignorant person, at a reasonable charge, and with a little bodily labour, might write books in philosophy, poetry, politics, laws, mathematics, and theology, without the least assistance from genius or study."[63] The machine was a silly device, a box of shafts and gears that spun through every possible combination of words, but it was a symbol of Jonathan Swift's mockery of the Royal Society. Babbage, though he shared some of Swift's reservations about the society, desired to avoid any comparison to the mythical device. He carefully described his invention in the context of de Prony's computing organization. He explained how the computers prepared their tables and then analyzed "what portion of this labour might be dispensed with." By his count, de Prony had employed ninety-six individuals to produce seven hundred computations a day. Babbage claimed that his machine could replace all of the human computers and most of the mathematicians. Of the original staff, all that would remain would be the ten planners and one mathematician, "or at the utmost two," to direct the work.[64]

Though the Royal Society was in no hurry to pass judgment upon Babbage's proposal, the Astronomical Society rushed to give the idea uncritical praise. Based upon what they observed of the crude prototype, they offered Babbage a gold medal for his contribution to astronomy.[65] "The labour of computing equations with the pen would be immense, and liable to innumerable errors," wrote one member of the Astronomical Society, "but with the assistance of [Babbage's] machine, they are all deduced with equal facility and safety." He made a special effort to emphasize the general use of the machine, arguing that "astronomical tables of every kind are reducible to the same general mode of computation" and that the machine could even be applied to commercial tasks, such as the preparation of "Interest, Annuities, &c, &c, all of which are reducible to the same general principles."[66]

The Royal Society eventually endorsed the Difference Engine, and the English government offered to finance the construction of the machine. Eager to devote his entire energies to the project, Babbage resigned his office with the Astronomical Society in 1823. The work progressed more slowly than he would have wished, as both he and his mechanic needed to refine his design and improve their metalworking skills. The project

was interrupted by the death of Babbage's wife, Georgiana, an event that disrupted his life in a way that nothing else could have. The loss "left Babbage a changed man," observed biographer Anthony Hyman. There was "an 'inner emptiness' to the man, who had only recently seen so much potential in his life." In "his public controversies, there was a new note of bitterness of which there was no trace while Georgiana was alive." Babbage left England for a Continental tour in 1827, leaving the engine unfinished.[67]

In later years, Babbage recognized that he had been naive in his attempt to build the Difference Engine. Without referring to himself directly, he confessed that a novice engineer could be "dazzled with the beauty of some, perhaps, really original contrivance," and would rush into its construction "with as little suspicion that previous instruction, that thought and painful labour, are necessary to its successful exercise."[68] Babbage worked on the Difference Engine for ten years. During this time, he was engaged in other projects, such as computing life insurance tables, forming two new scientific societies, and writing a book about manufacturing. Even accounting for these other projects, the work on the Difference Engine took longer than Babbage had anticipated, and it encountered unforeseen problems. The English government eventually grew impatient with Babbage's progress. Concluding that they would see no return on their investment, they withdrew their financial support in 1834, forcing Babbage to terminate the project.

Undeterred by his failure to complete the Difference Engine, Babbage moved to design a second, more ambitious device. He never even attempted to construct this machine, which he called the Analytical Engine. Modern writers have generally viewed the machine as an important intellectual step toward the stored-program electronic computer. One historian has gone so far as to claim that it was "a general purpose computer, very nearly in the modern sense."[69] The drawings of this machine show how Babbage anticipated the features of a modern computer, though his design used gears and levers rather than chips and circuit boards. The Analytical Engine had a means of storing numbers, a central processor, and an elementary programming mechanism. Unlike the Difference Engine, this machine was not restricted to a single mathematical method, such as the method of finite differences. The programming mechanism, which read instructions from a string of punched cards, controlled the order of operations. One observer, the daughter of the poet Lord Byron, Ada Lovelace (1815–1852), called the Analytical Engine the "material and mechanical representative of analysis," a triumph of the division of mathematical labor. Lovelace herself illustrated the nature of the machine by writing a sample program for it.[70]

Babbage would spend almost fifteen years designing the Analytical Engine. He left nearly three hundred detailed engineering drawings of his proposed machine.[71] As he worked over these drawings, he recognized that the Europe of the early nineteenth century might not be able to support large computing organizations or computing machines based upon the division of labor. The "most perfect system of the division of labour is to be observed," he wrote, "only in countries which have attained a high degree of civilization, and in articles in which there is a great competition amongst the producers."[72] As the early nineteenth century saw little competition for scientific computation, it offered little opportunity for the sophisticated division of labor espoused by Babbage and de Prony.